CSI 5328 Artificial Intelligence
Homework 3 (SVM)

- **Submission Deadline**:
  - 1st submission: October 19, 2026, 11:59 pm CST
  - 2nd submission: November 2, 2026, 11:59pm CST (**for correction only**)
- **Submission Format**: The submitted file must include
  - Student name;
  - Bear ID;
  - Course name and number;
  - Homework number.

  The title of the submitted file must include
  - Homework number;
  - Student name.

  An example of titles of submissions would be:
  - `HW#_Name.pdf` (for write-up assignments);
  - `HW#_Name.zip` (for coding assignments).
- Please submit your homeworks through **Canvas**. Homeworks sent to the instructor's email will **NOT** be accepted.
- Late Submission: -10 penalty for the first 24-hour; -20 penalty for $\geq$ 24 hour; Zero score for $\geq$ 3 days.
- If you have any questions, please send them to the instructor, Dr. Chen Zhao, at chen_zhao@baylor.edu.
- If you have any questions on your **grades**, please send them to the TA, Xiaohui Chen, at xiaohui_chen1@baylor.edu.

For all questions, your submission needs to include:

1. Your Python scripts for each question (*i.e.,* All files end with .py).
2. Screenshots of outputs in a **single** *pdf* file.

Question 1 [50pts]. You are given a dataset of two classes (0 and 1) represented in a 2D space. Your task is to use Support Vector Machines (SVM) to classify the data and visualize the decision boundary.

- Task 1 - Load the Data: Generate a synthetic dataset using the `make_blobs` function from `sklearn.datasets`. The dataset should contain two classes.
- Task 2 - Train-Test Split: Split the dataset into a training set (80%) and a test set (20%) using the `train_test_split` function from `sklearn.model_selection`.
- Task 3 - Train an SVM Classifier: Use the `SVC` class from `sklearn.svm` to create a linear SVM classifier. Train the classifier on the training data.
- Task 4 - Evaluate the Model: Evaluate the performance of your SVM model using accuracy. Predict the labels for the test set and calculate the accuracy using `accuracy_score`.
- Task 5 - Visualize the Decision Boundary: Plot the decision boundary of the trained SVM. Use the `plt.contourf` function to visualize the regions classified by the SVM.

Expected Output:

- A plot showing the synthetic dataset with two classes.
- The accuracy of the SVM on the test dataset, e.g., Accuracy: 95.00%.
- A plot showing the decision boundary created by the SVM classifier.

Question 2 [50pts]. You'll use the famous "Digits" dataset from `sklearn.datasets` to classify handwritten digits (0–9). This time, you'll test multiple SVM kernels and compare their performance.

- Task 1 - Load the Digits Dataset: The dataset consists of 8x8 pixel images of digits. You need to load it using the `load_digits` function from `sklearn.datasets`.
- Task 2 - Train-Test Split: Split the dataset into a training set (80%) and a test set (20%) using the `train_test_split` function from `sklearn.model_selection`.
- Task 3 - Experiment with Different Kernels: Train three SVM models with different kernels: 'linear', 'poly', and 'rbf'. Use the `SVC` class from `sklearn.svm` and compare the results.

- Task 4 - Evaluate and Compare the Kernels: For each kernel, calculate the classification accuracy. Which kernel performs the best? Use a table to present the results.
- Task 5 - Confusion Matrix: Plot the confusion matrix for the 'rbf' kernel to analyze where the model is making mistakes.

Expected Output:

- A few sample digit images will be displayed, so you know what you are dealing with (who doesn't love some visual flair?).
- After testing different kernels, you'll get accuracy scores like:
  - Linear Kernel Accuracy: 96%
  - Polynomial Kernel Accuracy: 97%
  - RBF Kernel Accuracy: 99%
- The confusion matrix for the RBF kernel should give you a detailed breakdown of correct and incorrect predictions for each digit.